

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	25 September 2026
Team ID	SWTID-2026-6223
Project Name	AI BlogNest API
Maximum Marks	4 Marks

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Technical Architecture:

The AI BlogNest API uses a client-API-database architecture. Client requests are received by the Node.js/Express.js API, application logic validates and processes requests, and MongoDB stores application data.

Layer / Component	Description	Technology
1. User Interface	Web/API client used to interact with the application and test endpoints.	HTML, CSS, JavaScript / Postman
2. Application Logic-1	Handles API routing, requests and responses.	Node.js
3. Application Logic-2	Provides REST API and middleware processing.	Express.js
4. Application Logic-3	Defines schemas, models and database operations.	Mongoose
5. Database	Stores users, authentication-related data and blog information.	MongoDB / NoSQL
6. Cloud Database	Cloud-hosted database service when deployed online.	MongoDB Atlas
7. File Storage	File storage is not required for the core API; local filesystem can be used if files are added later.	Local Filesystem / Other Storage
8. External API-1	External service integration is not required for the core BlogNest API.	Not applicable
9. External API-2	External service integration is not required for the core BlogNest API.	Not applicable
10. Machine Learning Model	No machine-learning model is specified in the provided project requirements.	Not applicable
11. Infrastructure (Server / Cloud)	Application runs on a local Node.js server during development and can be deployed to cloud infrastructure.	Local Server / Cloud deployment

Table-2: Application Characteristics:

S.No	Characteristics / Description	Technology
1	Open-Source Frameworks – Node.js and Express.js are used for backend API development; Mongoose is used for MongoDB interaction.	Node.js, Express.js, Mongoose

2	Security – Authentication, input validation, protected API operations, secure database credentials and environment-based configuration should be used.	Authentication, validation, environment variables, HTTPS in deployment
3	Scalable Architecture – Separation of client, API/application logic and database supports expansion of the application.	REST API, Node.js, Express.js, MongoDB
4	Availability – The API can be deployed on a reliable server/cloud environment with MongoDB Atlas for database availability.	Cloud deployment, MongoDB Atlas
5	Performance – Lightweight REST endpoints, JSON responses and efficient database queries support responsive API operations.	Node.js, Express.js, MongoDB

Note: The uploaded template lists example technologies and characteristics. The entries above are adapted specifically for the AI BlogNest API using the project's stated backend stack.